

A Readout Problem for Fefferman’s Existence and Smoothness of the Navier–Stokes Equation

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Core Epistemic Structure

Core Respondent / Experience-Based Expert: Yaoharee Lahtee — lived experience: sole author, originator of the admissible-domain-reading framework and the retained-turbulence continuation result; selections: chose the eight-state witness construction and the linear retained-ODE model as the minimal objects that make the readout-versus-domain distinction exact; interpretation: the Navier–Stokes readout dichotomy (R)/(N) below is posed, not answered, and the document’s own non-claim boundary (Remark 4) is authorial, not a hedge added after the fact.

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Non-collapse rule: Experience-Based Expertise $\neq$ Interactional Expertise $\neq$ AI Model. Registration is an attribution device, not a certification of truth or expertise.

Abstract

Fefferman’s note *Existence and Smoothness of the Navier–Stokes Equation* asks whether smooth, physically reasonable solutions of the three-dimensional Navier–Stokes equations exist globally, or whether breakdown can occur [1]. The present note does not alter that problem. It asks what additional statement is needed before a breakdown of the Navier–Stokes domain may be identified with a breakdown of a finite readout of an underlying discrete state. We prove one elementary finite-state theorem: exact commuting domain translation together with exact reader factorization does not imply that any finite-horizon reader decides an arbitrary domain question. We then pose, but do not answer, the corresponding Navier–Stokes readout dichotomy. As a further, genuinely separate result, we prove a global finite-interval continuation theorem for a finite retained-turbulence layer that could serve as one candidate finite reader: given finite dimension, positive time constant, a fixed finite generator, and a locally integrable drive, the retained state has a unique, bounded solution on every finite time interval. Consequently a hypothetical continuum breakdown does not, by itself, force blow-up of this retained layer, so long as its own finite-dimensional hypotheses continue to hold.

1 Continuum benchmark

Let $\Omega = \mathbb{R}^3$. For viscosity $\nu > 0$, consider

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad \nabla \cdot u = 0. \quad (\text{NS})$$

In Fefferman's formulation a physically reasonable solution is smooth for all time and, on $\mathbb{R}^3$, has uniformly bounded kinetic energy. Alternatives (A) and (B) are unforced global-existence statements; alternatives (C) and (D) allow a smooth external force and ask for breakdown [1]. Nothing below assumes that breakdown actually occurs; the breakdown predicate is only the domain-level event to be compared with finite readout events.

2 Finite state, domain map, and reader

We use a finite retained state $S_n = (G_n, \Lambda_n, T_n)$ and a finite stepper $S_{n+1} = F(S_n, u_n, c_n, T_n)$. For the argument below the controls, context and tape may be frozen, so we write simply $S_{n+1} = F(S_n)$.

Let D be a domain state space and let q_D map retained states into that domain. We call q_D an *admissible domain reading* when there are maps F_D and O_D such that

$$q_D(F(s)) = F_D(q_D(s)), \quad O(s) = O_D(q_D(s)). \quad (1)$$

The first identity says translation and evolution commute; the second says the declared reader is preserved by the translation.

For a nonnegative integer L , define

$$z \sim_{O,L} z' \iff O(F^k z) = O(F^k z') \text{ for every } 0 \leq k \leq L. \quad (2)$$

Two retained states may thus differ while remaining indistinguishable to a declared reader through a fixed finite horizon.

Let $Q : D \rightarrow \{0, 1\}$ be a yes-no question asked of the domain. We say O *decides* Q by time L if

$$z \sim_{O,L} z' \implies Q(q_D(z)) = Q(q_D(z')) \quad (3)$$

for all retained states z, z' . If no such L exists, every finite observation window leaves at least one pair of retained states that the reader identifies although the domain question separates them.

3 Exact finite non-entailment

Theorem 1 (Exact domain reading does not imply finite decision). *There is a finite retained system, an admissible domain reading satisfying (1), and a nonconstant domain question Q such that O does not decide Q by time L for any finite L .*

Proof. Take $S = \{0, 1\}^3$, $D = \{0, 1\}^2$, and define

$$F(a, b, c) = (1 - a, b, 1 - c), \quad F_D(a, b) = (1 - a, b), \quad (4)$$

$$q_D(a, b, c) = (a, b), \quad O(a, b, c) = b = O_D(a, b), \quad (5)$$

and $Q(a, b) = a$. Then

$$q_D(F(a, b, c)) = (1 - a, b) = F_D(q_D(a, b, c)), \quad O(a, b, c) = b = O_D(q_D(a, b, c)),$$

so (1) holds exactly on all 8 states, and $F^2 = \text{Id}_S$.

Now set $z = (0, 0, 0)$, $z' = (1, 0, 1)$. The second coordinate is fixed by F , hence $O(F^k z) = O(F^k z') = 0$ for every $k \geq 0$, so $z \sim_{O,L} z'$ for every finite L . The first coordinates remain complementary under every iterate of F , hence $Q(q_D(F^k z)) \neq Q(q_D(F^k z'))$ for every $k \geq 0$, and in particular $Q(q_D(z)) \neq Q(q_D(z'))$. Thus (3) fails for every finite L . $\square$

Corollary 2. *Exact domain commutation and exact readout factorization do not imply finite decision of every domain predicate.*

Remark 3. The theorem is not a statement about the Navier–Stokes equations. It is a finite separation result. Even when a domain translation commutes exactly with the dynamics and preserves the declared reader, a distinction made in the domain need not be a distinction made by that reader. This construction and its independent numerical re-verification are registered in the Toledo equation library as a proposal, pending canonical-code assignment, under the identifier PROP-NS-WITNESS-01 (tier `finite_diagnostic`: script-verified by exact enumeration over all 8 states, not machine-checked in Coq).

4 The Navier–Stokes readout problem

We now apply this observation only to the form of Fefferman’s question. Let D_{NS} denote a Navier–Stokes domain whose elements carry enough information to state Fefferman’s conditions. Let

$$\Sigma_F(d) = \begin{cases} 1, & \text{if } d \text{ has no global smooth solution satisfying Fefferman's conditions,} \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

This is only a shorthand for the breakdown property in alternatives (C) and (D) of [1]; it is not a new Navier–Stokes criterion.

To relate a finite retained system to this domain one must construct a translation $q_{NS} : S \rightarrow D_{NS}$ and prove, for the chosen domain dynamics and reader,

$$q_{NS}(F(s)) = F_{NS}(q_{NS}(s)), \quad O(s) = O_{NS}(q_{NS}(s)). \quad (7)$$

No such construction is supplied here. Its existence is part of the problem.

Assume such a translation has been fixed. The associated *Fefferman readout question* is whether the binary property Σ_F is determined by a finite prefix of the declared reader. More precisely, for a chosen class $C \subseteq S$, prove one of the following two statements.

(R) *Finite readout determination.* There is a finite L such that, for all $z, z' \in C$,

$$z \sim_{O,L} z' \implies \Sigma_F(q_{NS}(z)) = \Sigma_F(q_{NS}(z')). \quad (8)$$

In this case the declared finite reader determines the Fefferman breakdown property on C after finitely many steps.

(N) *Finite readout non-determination.* For every finite L there exist $z_L, z'_L \in C$ such that

$$z_L \sim_{O,L} z'_L, \quad \Sigma_F(q_{NS}(z_L)) \neq \Sigma_F(q_{NS}(z'_L)). \quad (9)$$

In this case no finite observation window of the declared reader decides the Fefferman breakdown property on C .

For fixed q_{NS} , O and C , (R) and (N) are complementary. The mathematical content is not the dichotomy itself, but the construction of a nontrivial Navier–Stokes translation satisfying (7), the declaration of a reader with specified finite information content, and the proof of one side of the dichotomy.

This formulation separates three questions that are otherwise easy to conflate. First, Fefferman asks whether a global smooth bounded-energy Navier–Stokes solution exists. Second, a domain translation asks whether a retained discrete dynamics can be read as a Navier–Stokes dynamics while

preserving the specified reader. Third, the readout problem asks whether the resulting breakdown property is determined by a finite observation history.

Theorem 1 shows that the third question is not answered by the second. Consequently, even an exact admissible domain reading does not by itself give the implication

$$\text{Navier–Stokes domain breakdown} \implies \text{finite reader breakdown.} \quad (10)$$

For the Navier–Stokes domain, that implication requires an additional theorem about the particular translation and the particular reader.

Nothing here contradicts Fefferman’s alternatives (A)–(D). If (C) or (D) is proved, the continuum breakdown statement stands. The readout problem begins only when one asks whether that domain-level breakdown is also a necessary failure of a specified finite information channel.

Remark 4. This note makes one proved finite statement (Theorem 1) and poses one Navier–Stokes adapter problem. It does not prove global regularity, does not prove blowup, and does not identify a continuum singularity with an artefact. Its purpose is to isolate the additional mathematical bridge required before those different claims may be treated as the same claim. The retained-turbulence continuation result of Section 5 answers a narrower, adjacent question — what happens to one *candidate* finite reader under one concrete regularity hypothesis — and likewise proves nothing about the continuum equation itself.

5 A finite retained-turbulence layer

One concrete candidate for the finite reader O above is a retained-turbulence layer. Let $r_R : D_{NS} \rightarrow \mathbb{R}^m$ be a retained-turbulence map and write $I_R(t) = r_R(q_{NS}(s(t)))$. The retained model is

$$\tau_R \dot{I}_R + L_R I_R = S_R + \eta_R, \quad \tau_R > 0, \quad I_R(t) \in \mathbb{R}^m, \quad L_R \in \mathbb{R}^{m \times m}. \quad (11)$$

Write $G_R(t) := S_R(t) + \eta_R(t)$. Equation (11) is not identified with (NS); it is a second state equation on a finite retained carrier. Its statement coincides exactly with the already-registered Toledo reading `weld/P.05.v1` (the URCF retained-information relaxation-inertia turbulence law); we cite that reading directly rather than re-registering the equation, and register only the new theorem below.

Theorem 5 (Global finite-interval continuation). *Assume*

$$m < \infty, \quad \tau_R > 0, \quad L_R \in \mathbb{R}^{m \times m} \text{ is fixed}, \quad G_R \in L^1_{loc}([0, \infty); \mathbb{R}^m). \quad (12)$$

Then for every $I_R(0) = I_0 \in \mathbb{R}^m$, equation (11) has a unique absolutely continuous solution on every finite interval, given by

$$I_R(t) = e^{-tL_R/\tau_R} I_0 + \frac{1}{\tau_R} \int_0^t e^{-(t-s)L_R/\tau_R} G_R(s) ds. \quad (13)$$

Moreover,

$$\|I_R(t)\| \leq e^{t\|L_R\|/\tau_R} \left(\|I_0\| + \frac{1}{\tau_R} \int_0^t \|G_R(s)\| ds \right), \quad (14)$$

hence $\sup_{0 \leq t \leq T} \|I_R(t)\| < \infty$ for every $T < \infty$.

Proof. Equation (11) is equivalent to $\dot{I}_R = -\tau_R^{-1} L_R I_R + \tau_R^{-1} G_R$. Variation of constants gives (13). Since $\|e^{-tL_R/\tau_R}\| \leq e^{t\|L_R\|/\tau_R}$, the norm bound follows from the triangle inequality. Local integrability of G_R makes the right-hand side finite for every finite t . $\square$

Corollary 6 (Positive-semidefinite restoration). *If $L_R = L_R^\top \succeq 0$, then $\|e^{-tL_R/\tau_R}\|_2 \leq 1$ and therefore*

$$\|I_R(t)\|_2 \leq \|I_0\|_2 + \frac{1}{\tau_R} \int_0^t \|G_R(s)\|_2 ds, \quad (15)$$

i.e. the exponential growth factor in Theorem 5's bound may be dropped to 1 whenever L_R is positive semidefinite.

Proof. Diagonalize $L_R = U\Lambda U^\top$ with $\lambda_j \geq 0$. Then $\|e^{-tL_R/\tau_R}\|_2 = \max_j e^{-t\lambda_j/\tau_R} \leq 1$. Applying this bound to (13) via the triangle inequality gives

$$\|I_R(t)\|_2 \leq \|e^{-tL_R/\tau_R}\|_2 \|I_0\|_2 + \tau_R^{-1} \int_0^t \|e^{-(t-s)L_R/\tau_R}\|_2 \|G_R(s)\|_2 ds \leq \|I_0\|_2 + \tau_R^{-1} \int_0^t \|G_R(s)\|_2 ds.$$

(The squared, cross-term-free bound $\|I_R(t)\|_2^2 \leq \|I_0\|_2^2 + \tau_R^{-1} \int_0^t \|G_R(s)\|_2^2 ds$ claimed in an earlier draft of this corollary does not follow from this argument and is false in general: $L_R = 0$, $\tau_R = 1$, $I_0 = 0$, $G_R \equiv 1$ on $[0, 2]$ gives $\|I_R(2)\|_2^2 = 4$ against a claimed bound of 2. The energy identity of Proposition 7 likewise yields, via Cauchy–Schwarz on $\langle I_R, G_R \rangle \leq \|I_R\|_2 \|G_R\|_2$, only this same linear bound, not a quadratic one.) $\square$

Proposition 7 (Energy identity). *Under $L_R = L_R^\top \succeq 0$,*

$$\frac{\tau_R}{2} \frac{d}{dt} \|I_R(t)\|_2^2 + \langle I_R, L_R I_R \rangle = \langle I_R, G_R \rangle. \quad (16)$$

Proof. Take the Euclidean inner product of (11) with I_R . $\square$

Theorem 8 (Failure localization). *Let $T_* > 0$ and define the retained regularity condition*

$$H_R(T_*) := \left[m < \infty, \tau_R > 0, L_R \in \mathbb{R}^{m \times m} \text{ fixed}, G_R \in L^1([0, T_* + \varepsilon]; \mathbb{R}^m) \text{ for some } \varepsilon > 0 \right], \quad (17)$$

and let $\Sigma_R(T_) := \mathbf{1}\{\limsup_{t \uparrow T_*} \|I_R(t)\| = \infty\}$. Then for every $T_* < \infty$,*

$$H_R(T_*) \implies \Sigma_R(T_*) = 0, \quad \text{hence} \quad \Sigma_F(T_*) \wedge H_R(T_*) \implies \Sigma_R(T_*) = 0. \quad (18)$$

Proof. The first implication is Theorem 5 applied on $[0, T_* + \varepsilon]$. The second follows because a hypothetical Navier–Stokes breakdown Σ_F does not by itself alter the hypotheses of the finite-dimensional continuation theorem. $\square$

Corollary 9 (Necessary bridge failure for transferred blow-up). *Any theorem of the form $\Sigma_F(T_*) \implies \Sigma_R(T_*) = 1$ requires $\Sigma_F(T_*) \implies \neg H_R(T_*)$. Thus at least one of the following must occur at breakdown: r_R ceases to be defined; L_R ceases to be a fixed finite operator; $G_R \notin L_{loc}^1$; τ_R ceases to remain positive; or the retained model is no longer the declared model.*

Remark 10. Theorem 5–Corollary 9 are registered in the Toledo equation library as a proposal, pending canonical-code assignment, under the identifier `PROP-NS-CONTINUATION-01` (tier `Dr`: a checkable analytical result, independently re-derivable, explicitly not `Th_coqc`), parented to the equation reading `weld/P.05.v1` (registered in Toledo's canonical registry, but itself carrying `status: unverified` and `coq_status: open_prop` for the underlying turbulence-physics claim – a fact this parenting does not inherit around, since the present theorem's own linear-ODE mathematics does not depend on that claim's physical verification) and to the retained-Laplacian root `EQ-008`. These results say nothing about the continuum equation (NS) itself. They say that *if* a candidate finite reader of the type (11) continues to satisfy its own finite-dimensional regularity hypotheses $H_R(T_*)$ up to a hypothetical Navier–Stokes breakdown time, that reader does not itself blow up. Whether H_R actually survives contact with a genuine Navier–Stokes translation q_{NS} satisfying (7) is precisely an instance of the unanswered dichotomy (R)/(N) of Section 4, not a fact established here.

6 Finite verification

The eight-state witness of Theorem 1 and the explicit-step stability of the retained stepper for equation (11) on a cycle-graph Laplacian ($m = 8$, $\lambda_{\max} = 4$, $\tau_R = 0.7$, explicit-step stability threshold $h \leq 2\tau_R/\lambda_{\max} = 0.35$) are checked directly by the script accompanying this paper (`verification/verify_state_breakdown_math.py`): exact enumeration over all 8 states confirms the weld identities and $F^2 = \text{Id}$; the witness pair is checked through $k = 9999$ steps; and a first-order explicit Euler stepper is compared against the exact matrix-exponential solution at four step sizes, giving error ratios converging to 2.0, consistent with $O(h)$ convergence for this finite test. These are finite, machine-checked facts about the objects defined above; they are not evidence about the continuum Navier–Stokes equation.

7 Conclusion

Exact domain-dynamics welding and exact reader factorization do not, by themselves, imply that a declared finite reader decides an arbitrary domain question (Theorem 1). Applied to Fefferman’s problem, this means an admissible Navier–Stokes domain translation does not by itself resolve whether the breakdown predicate Σ_F is finite-readout-determined; that is the Navier–Stokes readout problem, posed here as the dichotomy (R)/(N) and left open. Separately, one concrete candidate finite reader — a finite retained-turbulence layer governed by (11) — provably does not itself blow up on any finite interval so long as its own finite-dimensional hypotheses hold (Theorem 5), which sharpens *where* any future bridge theorem transferring continuum breakdown onto that reader would have to fail (Corollary 9). Neither result proves, disproves, or otherwise makes progress on the Clay Millennium Problem itself.

References

- [1] C. L. Fefferman, *Existence and Smoothness of the Navier–Stokes Equation*, in *The Millennium Prize Problems*, Clay Mathematics Institute / American Mathematical Society, 2006.